ZAINAB RAKIAH

Bangalore, Karnataka | +91 8861044384 | zainabrakiah2018@gmail.com | github.com/ZainabRakiah | linkedin.com/in/zainab-rakiah

PROFILE

CS undergraduate specializing in AI/ML with proven expertise in building scalable distributed systems, microservices architectures, and RESTful APIs. Demonstrates strong command of system design fundamentals, algorithms, and data structures. Experienced in designing fault-tolerant systems, optimizing performance at scale, and solving complex technical problems. Seeking SDE internship to contribute to production systems handling millions of requests while learning from experienced engineers.

EDUCATION

Ghousia College of Engineering, BANGALORE 2023 – 2027
B.E. Computer Science Engineering (Artificial Intelligence & Machine Learning) CGPA: 8.46

TECHNICAL SKILLS

Core CS Fundamentals: Object-Oriented Design (SOLID principles), Data Structures & Algorithms, System Design, Operating Systems, Computer Networks, Complexity Analysis

Backend & Distributed Systems: Microservices Architecture, RESTful API Design, Spring Boot, Node.js, Express.js, Scalable System Design, Fault-Tolerance Patterns, Load Balancing

Databases & Data Systems: MongoDB, PostgreSQL, MySQL, Query Optimization, Data Indexing, ACID Properties

Programming Languages: Java, Python, JavaScript

AI/ML & LLMs: LLM API Integration (OpenAI), Retrieval-Augmented Generation (RAG), Prompt Engineering, Machine Learning, Feature Engineering, Exploratory Data Analysis

Infrastructure: Git/GitHub, Microservices Deployment, CI/CD Fundamentals, AWS (Basics), Postman, Swagger

Testing & Quality Assurance: JUnit (Unit Testing), JBehave (End-to-End Testing), Test-Driven Development, Postman API Testing

PROJECTS

Restaurant Management System (github.com/ZainabRakiah)

Architected and built a scalable microservices-based food delivery platform using Spring Boot, demonstrating understanding of distributed systems design patterns and backend service decomposition.

Designed and implemented RESTful APIs across multiple service modules (User Management, Order Processing, Payment, Delivery) following REST conventions and API versioning best practices.

Implemented comprehensive test coverage using JUnit for unit tests and JBehave for end-to-end BDD testing, ensuring system reliability and correctness of business logic.

Utilized Git for version control and Gradle for automated build management, following Agile practices with sprint-based development cycles.

MapStop Holidays (github.com/ZainabRakiah)

Built an LLM-powered chatbot leveraging OpenAI API integration to handle travel planning queries, demonstrating proficiency with modern AI APIs and large-scale language model interactions.

Designed and implemented a Retrieval-Augmented Generation (RAG) pipeline to ground LLM responses in domain-specific travel data, improving factual accuracy and reducing hallucinations by ~40%.

Engineered multi-turn conversation handling with context preservation, optimizing prompt design for complex travel queries covering itineraries, destinations, and personalized recommendations.

Protego Emergency Response System with Predictive Safety Analytics (github.com/ZainabRakiah)

Performed large-scale data cleaning and preprocessing on 5,000+ emergency records, applying data structures knowledge for efficient data organization and implementing feature engineering for ML model training.

Designed and implemented grid-based spatial indexing system for fast geographic lookup, enabling nearest-hospital identification within < 100ms response time using efficient algorithms.

Built Decision Tree classifier for emergency prediction and safety recommendations, demonstrating understanding of algorithms and their practical application in production systems.

MERN Notes Application — Full-Stack CRUD System (github.com/ZainabRakiah)

Developed a full-stack JavaScript application using MERN stack (MongoDB, Express.js, React.js, Node.js) with complete CRUD functionality, demonstrating full-stack system design proficiency.

Designed scalable RESTful API architecture for note management with proper HTTP methods, status codes, and error handling patterns.

Integrated MongoDB for persistent data storage with optimized indexing for frequently queried fields, demonstrating database design and query optimization knowledge.

ACHIEVEMENTS

First Rank Holder in AIML Domain among 150+ students, Ghousia College of Engineering — Demonstrated exceptional performance in core CS fundamentals and system design coursework

Winner — Inter-College Hackathon (Opportunate) — Led team of 3 engineers from ideation to deployment, managing technical architecture decisions and coordinating cross-functional work

Runner-Up — Inter-College Debate Competition — Articulated complex technical concepts to non-technical audience, demonstrating communication skills essential for cross-team collaboration

Runner-Up — Katalyst India Technical Talk Competition — Presented technical deep-dive on emerging technologies to 200+ participants

LEADERSHIP & CONTRIBUTIONS

Led Opportunate hackathon team of 4 engineers; coordinated architecture design, sprint planning, and technical implementation from concept to deployment

Actively volunteered in 5+ hackathons and technical competitions, contributing to ideation, solution architecture, and implementation phases

Delivered technical presentations on AI, Edge Computing, IoT, and Machine Learning to peer groups and college forums, demonstrating ability to communicate complex ideas clearly

CERTIFICATIONS & CONTINUOUS LEARNING

Machine Learning Specialization (DeepLearning.AI by Andrew Ng) — Algorithms, neural networks, and practical ML engineering

Complete Python Bootcamp (Jose Portilla, Udemy) — Python programming fundamentals and data structures

Complete SQL Bootcamp (Jose Portilla, Udemy) — Database design, query optimization, and relational databases

Java Programming for Complete Beginners (Udemy) — Object-oriented design, design patterns, and backend development